

Package ‘illdthCIF1’

June 16, 2026

Title Cumulative Incidence Function Estimation in the Illness-Death Model

Version 1.1.1

Description R code for for applying to a dataset methods for estimating the cumulative incidence function for disease in an illness-death model.

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Encoding UTF-8

Roxygen list(markdown = FALSE)

RoxygenNote 8.0.0

Imports survival

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cifcmp.full	<i>Apply three methods (see note below) for estimating the cumulative incidence function for disease in an illness-death model.</i>
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Description

Apply three methods (see note below) for estimating the cumulative incidence function for disease in an illness-death model.

Usage

```
cifcmp.full(  
  age_recr,  
  age_diag,  
  age_death,  
  age_end_fu,  
  status_end,  
  tgrd,  
  tgrd.cb,  
  covpr,  
  nresam,
```

```

    dtgrd_width,
    bwselect,
    bwvec,
    nfl,
    redo_bw,
    verbosity
  )

```

Arguments

age_recr	age at recruitment
age_diag	age at diagnosis
age_death	age at death
age_end_fu	age at end of follow-up (death or censoring)
status_end	status at end of follow-up
tgrd	grid of ages at which the CIF will be computed
tgrd.cb	grid of ages over which the simultaneous confidence band will be computed (subset of tgrd)
covpr	desired confidence interval coverage probability
nresam	number of bootstrap replications for confidence band
dtgrd_width	grid width for age at diagnosis in conditional survival calculation
bwselect	TRUE/FALSE flag for whether to carry out bandwidth selection
bwvec	vector of candidate bandwidths
nfl	number of folds in crossvalidation procedure for selecting the bandwidth
redo_bw	TRUE/FALSE flag for redoing bandwidth selection in bootstrap
verbosity	code for verbosity level (0=none, 1=intermediate, 2=full)

Value

a list containing the estimates, pointwise confidence limits, and simultaneous confidence band limits for all three methods and the bandwidth parameter for the Zucker-Gorfine method

Note

Methods provided:

1. The method of Aalen and Johansen (1978)
2. The method of Gorfine, Zucker, and Shoham (2025)
3. The method of Zucker and Gorfine (2026)

Method #3 includes all disease cases.

status_end = status at end of follow-up
 0 = alive without disease (i.e. censored)
 1 = died without disease
 2 = alive with disease
 3 = died with disease

When no bandwidth selection is done, the bandwidth used is bwvec[1].

References

- Aalen, O. O. and Johansen, S. (1978). An empirical transition matrix for non-homogeneous Markov chains based on censored observations. *Scandinavian Journal of Statistics*, 5:141–150.
- Gorfine, M., Zucker, D. M., and Shoham, S. (2025). Cumulative incidence function estimation using populationbased biobank data. *Biometrics*, 81(3):ujaf049.
- Zucker, D. M. and Gorfine, M. Cumulative incidence function estimation in the illness-death model using all disease cases. *arXiv*.

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